Austica Memorial College, Nanka

2026 Summer Coding & Robotics Bootcamp

Programme Overview

Detail	Description
Theme	Learn • Build • Create • Innovate
Duration	4 Weeks (Monday – Friday)
Format	Hands-on practical bootcamp

Target Participants

The bootcamp is structured into three distinct class levels to ensure age-appropriate learning outcomes:

- **Foundation Class** — JSS1 – JSS2
- **Intermediate Class** — JSS3 – SS1
- **Advanced Class** — SS2 – SS3

Daily Schedule

Time	Activity
9:00 – 9:20	Morning Assembly / Ice Breaker
9:20 – 10:00	Theory Session
10:00 – 11:00	Practical Session
11:00 – 11:20	Break

Time	Activity
11:20 – 12:40	Project Work
12:40 – 1:00	Daily Review & Challenge

Common Activities (All Classes)

Every student participates in a shared set of activities that foster collaboration, creativity, and real-world problem solving:

- Team Building
- Coding
- Robotics
- Digital Skills
- Problem Solving
- Innovation Challenge
- Weekly Quiz
- Presentation Skills
- Project Exhibition

FOUNDATION CLASS (JSS1 – JSS2)

Goal

Introduce students to computers, programming, electronics, and simple robotics in a fun and engaging way that builds curiosity and foundational confidence.

Week 1 – Digital Skills & Computational Thinking

Day	Topic
Monday	Orientation, Bootcamp Rules, Technology Careers
Tuesday	Computer Basics (Hardware & Software)

Day	Topic
Wednesday	Keyboard Skills, File Management
Thursday	Computational Thinking & Algorithms
Friday	Blockly Games / MakeCode Arcade Activities

Project Highlights

- Interactive Maze
- Animation
- Logic Puzzle

Week 2 – Coding Fundamentals

Day	Topic
Monday	Programming Concepts
Tuesday	Events & Sequences
Wednesday	Variables
Thursday	Loops & Conditions
Friday	Mini Coding Challenge

Project Highlights

- Quiz Game
- Catch Game
- Number Game

Week 3 – Robotics & Electronics

Day	Topic
Monday	Introduction to Electronics
Tuesday	LEDs & Resistors
Wednesday	Breadboard & Circuits

Day	Topic
Thursday	Arduino Introduction
Friday	LED & Buzzer Project

Project Highlights

- Traffic Light
- Doorbell
- Flashing Lights

Week 4 – Creativity & Innovation

Day	Topic
Monday	Team Project Planning
Tuesday	Project Development
Wednesday	Testing
Thursday	Presentation Practice
Friday	Project Exhibition

INTERMEDIATE CLASS (JSS3 – SS1)

Goal

Develop programming skills and introduce embedded systems through practical projects that strengthen logical thinking and problem solving.

Week 1 – Python Programming

Day	Topic
Monday	Python Installation & First Program
Tuesday	Variables & Data Types

Day	Topic
Wednesday	Input & Output
Thursday	If Statements
Friday	Loops

Project Highlights

- Calculator
- Grade Checker
- Guessing Game

Week 2 – Python Projects

Day	Topic
Monday	Lists
Tuesday	Functions
Wednesday	File Handling
Thursday	Error Handling
Friday	Mini Project

Project Highlights

- Student Record System
- Contact Book
- Quiz Application

Week 3 – Robotics

Day	Topic
Monday	Arduino IDE
Tuesday	LEDs & Buttons
Wednesday	Buzzers

Day	Topic
Thursday	Sensors
Friday	Servo Motors

Project Highlights

- Traffic Light
- Smart Light
- Parking Sensor

Week 4 – Innovation Project

Day	Topic
Monday	Design Thinking
Tuesday	Team Project
Wednesday	Testing
Thursday	Presentation
Friday	Exhibition

ADVANCED CLASS (SS2 – SS3)

Goal

Prepare students for tertiary STEM education by combining software development, robotics, web development, and innovation into portfolio-ready projects.

Week 1 – Advanced Python

Day	Topic
Monday	Python Review
Tuesday	Functions

Day	Topic
Wednesday	Modules
Thursday	File Handling
Friday	Mini Project

Project Highlights

- Expense Tracker
- Library System
- Inventory Manager

Week 2 – Robotics & Embedded Systems

Day	Topic
Monday	Arduino Programming
Tuesday	Digital & Analog Inputs
Wednesday	Sensors
Thursday	LCD Display
Friday	Automation Project

Project Highlights

- Smart Door
- Water Tank Alarm
- Smart Classroom Light

Week 3 – JavaScript & Web Development 🆕

Goal: Introduce students to modern web development with JavaScript — covering both **frontend** (what users see and interact with) and **backend** (the server and database behind the scenes). By the end of the week, students will be able to build a simple full-stack web application.

Day	Topic
Monday	Introduction to the Web (HTML, CSS & JavaScript Overview)
Tuesday	JavaScript Basics (Variables, Data Types, Operators, Functions)
Wednesday	Frontend Development (DOM, Events, Forms & Interactivity)
Thursday	Backend Development (Node.js, npm & Express.js)
Friday	Connecting Frontend to Backend (APIs & JSON)

Project Highlights

- Interactive Quiz Web App
- Personal Portfolio Web Page
- Simple Full-Stack App (e.g., Student Directory)
- Digital Clock & Calculator
- School Notice Board Web App

Week 4 – Capstone Project (Full-Stack Web Application)

Day	Topic
Monday	Project Planning & UI Design
Tuesday	Frontend Development
Wednesday	Backend Development & Database Integration
Thursday	Testing, Debugging & Presentation Practice
Friday	Innovation Fair & Project Demonstration

Capstone Examples (Student Choice)

- School Result Management System
- Student Attendance Tracker Web App
- Class Blog with Comment System
- Online Quiz / CBT Platform
- Library Catalogue Web App

Friday Activities (All Classes)

Every Friday is designed as a showcase and motivation day. The following activities are run across all classes:

- Weekly Quiz
- Coding Challenge
- Robotics Challenge
- Team Competition
- Guest Speaker or Career Talk
- Project Demonstration
- Awards (Best Team, Best Coder, Best Builder, and other recognitions)

Expected Outcomes

By the end of the bootcamp, students will have:

- Built functional coding and robotics projects appropriate to their level
- Developed presentation and teamwork skills
- Explored emerging technologies (AI, IoT, automation)
- **Advanced students:** Built a full-stack web application using JavaScript (frontend + backend)
- Demonstrated learning through a final exhibition or innovation fair

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Prepared for: Austica Memorial College, Nanka — 2026 Summer Programme